

Prime Numbers: An integer $p > 1$ is called a prime number, or simply a prime, if its only positive divisors are 1 and p .

Ex: 2, 3, 5, 7, 11 etc

An integer greater than 1 that is not a prime is termed composite.

Ex: 4, 6, 8, 9, 10 etc.

Note: 2 is the only even prime.

1 is neither prime nor composite.

Theorem: If p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: If $p \mid a$, then we no need to prove.

Assume that $p \nmid a$.

$\Rightarrow \gcd(p, a) = 1$. ($\because$ the only positive divisors of p are 1 and p itself)

$\Rightarrow$ Hence, $p \mid b$, by Euclid's lemma

Ex: $3 \nmid 42 = 6 \times 7$ and $3 \nmid 6$ even though $3 \nmid 7$.

Whereas $4 \nmid 12 = 2 \times 6$, but $4 \nmid 2$ as well as $4 \nmid 6$ (as 4 is not a prime)

Note: 1) If p is a prime and $p \mid a_1 a_2 \dots a_n$, then $p \mid a_i$, for some $1 \leq i \leq n$.

2) If $p, q_1, q_2, \dots, q_n$ are all primes, and $p \mid q_1 q_2 \dots q_n$, then

$$p = q_i, \text{ for some } 1 \leq i \leq n.$$

Fundamental Theorem of Arithmetic.

Every positive integer $n > 1$ is either a prime or a product of primes; this representation is unique, apart from the order in which the factors occur.

Proof:- If n is prime, then $n = n$ is the required representation.

If n is a composite number, then n has a non-trivial divisor, say a .

$\Rightarrow a | n$ or $n = ab$, for some integer b .

Now, $n = ab$ and $1 < a < n$
 $1 < b < n$.

It is enough, if we prove the theorem for a and b , instead of n .

Repeating the same argument for a , b and smaller

factors, which are all greater than 1, but less than n ,
 we get primes, $p_1, p_2, \dots, p_k$
 such that $n = p_1 p_2 \dots p_k$.

Uniqueness:-

Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ and

$n = q_1^{\beta_1} q_2^{\beta_2} \dots q_r^{\beta_r}$ be two

distinct representations of n .

Then $p_i \mid n = q_1^{\beta_1} \dots q_r^{\beta_r}$, $\forall i$

$\Rightarrow p_i = q_j$ for some j , $1 \leq j \leq r$.

Now $q_j^{\beta_j} = p_i^{\beta_j}$ and $p_i^{\alpha_i} \mid n$

$$\Rightarrow p_i^{\alpha_i} \mid p_i^{\beta_j}$$

$$\Rightarrow \alpha_i \leq \beta_j$$

For the similar argument,

$$\beta_j \leq \alpha_i$$

$$\Rightarrow \alpha_i = \beta_j \text{ whenever } p_i = q_j$$

$$\Rightarrow r = s$$

$$\therefore n = p_1^{\alpha_1} \dots p_r^{\alpha_r} \text{ and}$$

$$n = q_1^{\beta_1} \dots q_s^{\beta_s} \text{ are same}$$

representations, except the order of primes

$\Rightarrow$ Uniqueness.

Canonical representation:-

For $n > 1$, the representation $n = p_1^{\alpha_1} \dots p_r^{\alpha_r}$, with $p_1 < p_2 < \dots < p_r$ is a canonical representation of n as a product of prime numbers.

Ex:- $4200 = 2^3 \times 3 \times 5^2 \times 7$

$$10780 = 2^2 \times 5 \times 7^2 \times 11.$$

Q.C.D:-

$$\text{If } a = p_1^{\alpha_1} \dots p_n^{\alpha_r}$$

$$b = p_1^{\beta_1} \dots p_n^{\beta_r}$$

$$\text{then } \text{g.c.d.}(a, b) = p_1^{\gamma_1} \dots p_r^{\gamma_r}$$

$$\text{Show each } \gamma_i = \min\{\alpha_i, \beta_i\}$$

$$\begin{aligned} \underline{\text{Ex:-}} \quad \text{gcd}(4200, 10780) \\ = 2^2 \times 5 \times 7 = \underline{\underline{140}} \end{aligned}$$

Exactly divides:-

$p^\alpha \parallel a$ (i.e., p^α exactly divides a)

if $p^\alpha \mid a$ and $p^{\alpha+1} \nmid a$.

(i.e., α is the largest exponent

$\exists p$, such that $p^\alpha | a$

EX:- $2^3 || 24$ ($\because 2^3 = 8 | 24$
but $2^4 = 16 \nmid 24$)

Note:- ① If $n = p_1^{\alpha_1} \dots p_r^{\alpha_r}$ is the
canonical representation of n ,
then $p_i^{\alpha_i} || n$.

② If $n = p_1^{\alpha_1} \dots p_r^{\alpha_r}$, then
no. of divisors of n is equal
to $(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_r + 1)$.

(In fact, $p_i^{\beta_k} | n$, for $1 \leq k \leq \alpha_i$,
and $\nmid n$).

$\Rightarrow$ there are $\alpha_i + 1$ choices for

each i , which are $p_i^0, p_i^1, p_i^2, \dots, p_i^{s_i}$

Ex:- $24 = 2^3 \times 3$

$\Rightarrow$ There are $(3+1)(1+1) = 8$
divisors of 24.

They are: 1, 2, 3, 4, 6, 8, 12, 24

Ex:- Prove that any prime of
the form $3n+1$ is also of the
form $6m+1$.

Let $p = 3n+1$ is a prime.

$\Rightarrow p$ is odd. $\Rightarrow p-1 = 3n$ is even.

$\Rightarrow n$ is even, let $n = 2m$.

$\Rightarrow p = 3(2m) + 1 = \underline{\underline{6m+1}}$

Ex:- Each integer of the form $3n+2$ has a prime factor of this form.

Let $3n+2 = p_1 p_2 \dots p_k$ be the factorization (allowing the repetition)

Each p_i is of the form, $3q_i+1$ or $3q_i+2$ (by division algorithm).

If all $p_i = 3q_i+1$, then

$$p_1 p_2 \dots p_k = 3l+1 \neq 3n+2$$

(by uniqueness of division algorithm).

$\therefore$ there must be at least one prime factor of the form $3n+2$

Ex:- Prove that- only prime of the form n^3-1 is 7.

$$n^3-1 = (n-1)(n^2+n+1)$$

If n^3-1 is prime either

$n-1=1$ or $n^2+n+1=1$, which is not possible.

$$\Rightarrow n=2$$

$$\Rightarrow \underline{\underline{n^3-1=7}}$$

Ex:- Prove that- the only prime p for which $3p+1$ is a perfect square is $p=5$.

$$3p+1 = n^2 \Rightarrow 3p = n^2-1 = (n-1)(n+1)$$

$$\text{If } p=5, \text{ then } 3p+1=16=4^2 \\ \Rightarrow \underline{\underline{n=4}}$$

$$3p = (n-1)(n+1) \Rightarrow n-1=3, n+1=p \\ \text{or}$$

$$n-1=p, n+1=3$$

$$\Rightarrow \underline{n=4} \text{ or } \underline{n=2}$$

$$n=2 \Rightarrow 3p+1=2^2=4$$

$$\Rightarrow 3p=3 \Rightarrow p=1 \text{ (not possible)}$$

$p=5$ is the only case.

EX1:- The only prime of the form n^2-4 is 5.

$$p=n^2-4=(n-2)(n+2)$$

$$\Rightarrow \begin{cases} n-2=1 & \& n+2=p \\ \text{OR} \\ n-2=p & \& n+2=1 \end{cases}$$

$$\Rightarrow \begin{cases} n=3 & \text{and } p=5 \\ \text{OR} \\ n=-1 & \text{and } p=-3, \\ & \text{not possible.} \end{cases}$$

$\therefore \underline{p=5}$ is the only prime.

Ex:- A prime $p|a^n \Rightarrow p^n|a^n$.

$$p|a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ times}}$$

$$\Rightarrow p|a \Rightarrow a = p^k$$

$$\Rightarrow a^n = p^n k^n$$

$$\Rightarrow \underline{p^n|a^n}$$

Ex:- If $n > 4$ is a composite, then
 $n|(n-1)!$

$$\text{Let } n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$$

$$\text{If } r > 1, \text{ then } p_i^{\alpha_i} | n, \forall i$$

$$\Rightarrow p_i^{\alpha_i} < n \Rightarrow p_i^{\alpha_i} \leq n-1$$

$$\Rightarrow p_i^{\alpha_i} | (n-1)!, \forall i$$

$$\Rightarrow \prod p_i^{\alpha_i} = n | (n-1)!$$

Suppose, $n=1 \Rightarrow n=p^\alpha$

n is composite $\Rightarrow \underline{\underline{\alpha \geq 1}}$

$$\Rightarrow n = p \cdot p^{\alpha-1}$$

$$\Rightarrow n > p \text{ and } n > p^{\alpha-1}$$

$$\Rightarrow n-1 \geq p \text{ and } n-1 \geq p^{\alpha-1}$$

If $p \neq p^{\alpha-1}$, then both p and $p^{\alpha-1}$ are in the representation of $(n-1)!$

$$\therefore p \cdot p^{\alpha-1} = n \mid (n-1)!$$

If $p = p^{\alpha-1}$, then $\alpha-1=1$ or $\alpha=2$.

$$\Rightarrow n = p^2$$

But $n > 4 \Rightarrow p \neq 2$.

$$n \geq 6 \Rightarrow 2(n-1) < (n-1)!$$

$\Rightarrow 2(n-1)$ is in terms of $(n-1)!$

$\therefore p$ and $2p$ are in terms of $(n-1)!$

$$\Rightarrow p \cdot 2p \mid (n-1)!$$

$$\Rightarrow 2p^2 \mid (n-1)! \quad \Rightarrow p^2 = n \mid (n-1)!$$

Gaussian integers:-

$\alpha = n + mi$, (where n, m are integers)
are Gaussian integers.

Ex:- Use Euclidean algorithm to

find g.c.d. $(5+6i, 3-2i)$

$$|5+6i| = \sqrt{5^2+6^2} > \sqrt{3^2+(-2)^2} = |3-2i|$$

$$\frac{5+6i}{3-2i} = \frac{(5+6i)(3+2i)}{3^2+2^2}$$

$$= \frac{1}{13} (3 + 28i)$$

$$\approx 0 + 2i$$

$\frac{3}{13}$ is closer to 0 than 1

$\frac{28}{13}$ is closer to 2 than 3

Note:-

$$\frac{3}{13} = 0.2307 < 0.5$$

$$\frac{28}{13} = 2.1538 \approx 2.$$

$$\therefore 5 + 6i = \underbrace{2i(3 - 2i)}_{4 + 6i} + 1$$

$$= 2i(3 - 2i) + (1)$$

$$\therefore \text{g.c.d.}(5 + 6i, 3 - 2i) = 1.$$

Moreover,

$$\underline{\underline{1 = (5 + 6i) \times 1 - 2i(3 - 2i)}}$$

Ex:- Find g.c.d. $(7-11i, 8-19i)$

$$8-19i = 2(7-11i) + (-6+3i)$$

$$7-11i = (-2+i)(-6+3i) + (-2+i)$$

$$-6+3i = 3(-2+i) + 0.$$

$$\therefore \text{g.c.d.}(7-11i, 8-19i) \\ = \underline{\underline{-2+i}}$$

Moreover,

$$-2+i = (-3+2i)(7-11i) \\ + (2-i)(8-19i) \\ \underline{\underline{\hspace{1cm}}}$$

Note:-

Gaussian integers are complex numbers with real and imaginary parts are integers.

They are the vertices of the

squares of grid.

If α and β are Gaussian integers, then $\alpha \mid \beta$ if there is a Gaussian integer γ such that $\beta = \alpha\gamma$.

$\text{g.c.d.}(\alpha, \beta) = \delta$, where δ is a Gaussian integer of maximum absolute value which divides both α and β .

Note:- g.c.d. of Gaussian integers is not unique, as by multiplying ± 1 and $\pm i$, we get Gaussian integers with same absolute value and dividing both α and β .

Ex:- If $p \mid b^6 + 1$, where p is a prime and $b^6 + 1$ is an integer, then p can be expressed as $p = c^2 + d^2$, for some integers c and d .

In fact, $b^6 + 1 = (b^2 + 1)(b^4 - b^2 + 1)$

If $p \mid b^6 + 1$, then,

$$p \mid b^2 + 1 \quad \text{or} \quad p \mid b^4 - b^2 + 1.$$

① If $p \mid b^2 + 1 = (b+i)(b-i);$

let $c+di = \text{g.c.d.}(p, b+i).$

Then $p = (c+di)(c-di)$

$$\Rightarrow \underline{\underline{p = c^2 + d^2}}$$

② If $p \mid b^4 - b^2 + 1 = (b^2 - 1)^2 + b^2$

$$\Rightarrow p \mid [(b^2 - 1) + bi][(b^2 - 1) - bi]$$

$$\text{Let } \gcd(p, (a^2-1)+bi) = c+di$$

$$\Rightarrow p = (c+di)(c-di)$$

$$\Rightarrow \underline{\underline{p = c^2 + d^2}}$$

Ex:- If $12277 \mid 20^6 + 1$, find
express the prime 12277
as a sum of two squares.

$$\text{Ans: } \underline{\underline{12277 = 89^2 + 66^2}}$$

Ex:- $769 \mid 19^6 + 1 \Rightarrow$ Express 769
as a sum of two squares.

